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RESEARCH ARTICLE

Can information and advising affect postsecondary participation and attainment for military personnel? Evidence from a large-scale experiment with the U.S. Army

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Abstract
Despite generous financial aid, military veterans have high rates of undermatch and generally poor postsecondary outcomes. We conducted a large-scale, multi-arm field experiment with the U.S. Army to investigate whether personalized information about postsecondary options and access to advising affects service members' postsecondary choices and outcomes. We find no impact of the intervention on whether or where veterans enroll in college or on their college persistence. These results suggest that light touch strategies that have been effective at addressing similar challenges among traditional students, and which we modified for the military context, are not sufficient to improve veterans' postsecondary outcomes.

INTRODUCTION

For over a decade, policymakers and veterans advocates have expressed concern about the postsecondary choices and outcomes of U.S. military veterans. While military service appears to increase rates of postsecondary participation, veterans enroll disproportionately at less selective institutions and are unlikely to attain bachelor's degrees (Greenberg et al., 2022; Loughran et al., 2011). Compounding these concerns, veterans' worse postsecondary outcomes occur despite substantial public investment and policy interest in supporting their postsecondary educational success. Recent GI Bill expenditures exceed $10 billion annually or approximately $17,400 per student served, more than 3 times the generosity of the Federal Pell Grant (Bass, 2019; Office of Postsecondary Education, 2018).

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One reason for poor postsecondary outcomes among veterans is their comparatively high rates of enrollment at institutions with low rates of student success. For example, 24% of enrolled veterans attend for-profit schools, similar to rates among other non-traditional student populations (e.g., 18% of independent students), but much higher than traditional students (e.g., 5% of dependent students).[1] For-profit institutions typically have credentials with lower labor market value and generally worse student outcomes (Armona et al., 2022; Cellini & Turner, 2019; Darolia et al., 2015; Government Accountability Office [GAO], 2019). Accompanying the high rate of for-profit attendance is substantial academic undermatch among veterans: Within the population that scores in the top decile of a nationally normed aptitude test, service members are less than half as likely to attend selective institutions than their civilian counterparts (33% versus 73% attendance rate, respectively; Barr et al., 2019).

One factor contributing to observed undermatch may be veterans' lack of awareness of the importance of college quality or their potential admissibility to selective institutions. Veterans may also have lower familiarity with the college process (e.g., many lack parents or peers with college degrees) and have more limited access to education planning resources relative to traditional-age students in structured high school settings. Another factor may be that service members overestimate the net price they would pay to attend selective colleges and universities (Avery & Kane, 2004; Grodsky & Jones, 2007; Horn et al., 2003), or that service members are deterred by the perceived complexity of the process of applying for and receiving GI Bill benefits (Bettinger et al., 2012; Dynarski & Scott-Clayton, 2006).

College choice frictions for veterans are further exacerbated by deceptive and manipulative recruiting practices used by for-profit institutions to entice veterans to enroll at their institutions (Committee on Health, Education, Labor, and Pensions [Committee on Health], 2012). In the absence of alternative information and assistance, veterans may be swayed by the misleading and false information that recruiters provide about program lengths and costs, and about job placement rates and salaries among graduates.[2]

Social psychological barriers, including feelings of belonging and stereotype threat, may also impede military personnel from applying to or attending selective colleges or universities (Aronson et al., 2002; Walton & Cohen, 2007, 2011). Finally, service members may intend to apply to or attend selective institutions but miss important deadlines or processes because of limited attention or procrastination (Castleman & Page, 2015; Karlan et al., 2016).

We designed an intervention to address these frictions, drawing on insights and strategies used to address similar frictions among traditional student populations. Prior evidence has demonstrated, for instance, that providing high-achieving, low-income students with customized information about well-matched colleges or clarifying that an individual will pay nothing to attend a selective school can lead to substantial improvements in enrollment quality (Dynarski et al., 2021; Hoxby & Turner, 2013). Providing students personalized assistance with college- and financial aid–related tasks can increase college enrollment (Bettinger et al., 2012; Castleman & Page, 2015).[3] Social psychological research has found that addressing anxieties students have about whether they belong in college can improve academic performance and physical health (Walton & Cohen, 2011). Finally, individualized college

[1] Authors, calculations from 2015:16 NPSAS.

[2] The investigation moreover documented explicit efforts by for-profit institutions to target military service members by locating recruiters near installations; placing ads in high-circulation military publications; and conducting direct outreach through phone, email, and web advertising. Recruiters went so far as to seek out veterans at military hospitals and wounded warrior rehabilitation centers (Committee on Health, 2012). More recently, out of continued concern over for-profit institutions recruitment practices, the Biden administration closed the "90/10 loophole" to reduce these institutions' incentives to target veterans (Shane, 2022; Weissman, 2021).

[3] More recent experimental studies have found smaller or no effects of similar informational and nudge strategies focused on providing traditional high-school-age students with information about college quality (e.g., Gurantz et al., 2019). One potential explanation for this result is the more limited extent of undermatch in more recent studies (i.e., many of the students were already attending institutions with similar average SAT scores in the absence of the intervention). A related possibility noted by Gurantz et al. (2019) is that the extent of outreach among control group students likely increased significantly after earlier studies raised awareness of the extent of mismatch among high-achieving traditional students. Of course there are other more specific reasons why each particular intervention may have been ineffective. It is worth noting that the results of many of these recent studies were not available when we designed our intervention.

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advising—particularly programs that focus on encouraging students to apply to well-matched colleges and universities—can generate large improvements in enrollment quality and degree attainment (Barr & Castleman, 2021).

We conducted a large-scale randomized control trial to investigate whether adapting similar strategies to the U.S. Army context could help overcome frictions in the choice process and improve postsecondary outcomes among military personnel. In order to increase service members' awareness of well-matched, affordable colleges and universities, and to support them to make informed decisions about whether and where to pursue postsecondary education, we provided enlisted service members separating from the U.S. Army with personalized and simplified information, reminders, and advising about their college and university options. We specifically focused on encouraging service members to apply to *well-matched, selective, and affordable* colleges and universities because a growing body of research demonstrates that improvements to the quality of institution at which students enroll lead to both higher rates of attainment and better labor market outcomes (Angrist et al., 2022; Barr & Castleman, 2021; Goodman et al., 2017; Smith et al., 2020; Zimmerman, 2014). For every treated service member, we generated a personalized list of four colleges close to where the service member planned to live after leaving the Army; we also created a project-specific website where treated service members could generate an extended list of recommended colleges for any location in the country.[4]

In addition to providing personalized information on well-matched colleges and universities, our intervention materials incorporated several other evidence-based strategies for improving postsecondary outcomes among traditional-age student populations. Specifically, to address concerns service members might have about college affordability, our materials conveyed to service members that they would face a net price of $0 (net of GI Bill Funding) at most colleges and universities *without* having to compromise on college quality (Dynarski et al., 2021).[5] Our materials moreover leveraged the social belonging literature (e.g., Walton & Cohen, 2011) by including information about student veterans' groups at each of the institutions we identified and by providing contact information for group leaders, in order to address veterans' potential concerns about "fitting in" among a traditional student population. Given service members extended absence from academic environments, we also addressed anxiety about college entrance exam taking (e.g., Aronson et al., 2002; Jenner, 2017). Finally, to account for the numerous service-related demands on military personnels' time, we provided reminders for service members to follow through on important college application actions (e.g., Castleman & Page, 2015).

We delivered this information in multiple messages over a period of approximately 1 month through a combination of postal mail, email, text message, and our project website. In Figure 1 we present an overview of the intervention materials.[6] All treated service members (two-thirds of the experimental sample) were sent these materials. We also randomly assigned half of treated service members to receive additional proactive and personalized, text-based college advising, given prior evidence that interactive, text-based advising can improve postsecondary outcomes among traditional student populations (Castleman & Page, 2015, 2017; Page et al., 2020).[7]

We conducted our experiment in 2016 and 2017 with enlisted active-duty service members who were in the process of separating from six of the largest Army installations in the country ($N =$

[4] The average graduation rate for the institutions we identified in the project materials is at the 60th percentile in the national distribution of graduation rates at 4-year institutions; comparatively, the average graduation rate among institutions attended by service members in the absence of the intervention is at the 41st percentile. We calculate the latter based on the average graduation rate of institutions attended by control group service members in our experimental sample.

[5] Graduation rates, our proxy for quality, correlate strongly with measures of instructional expenditures, post-college earnings, loan repayment, and even estimates of value added (Rothwell & Kulkarni, 2015).

[6] We present the full set of materials in Appendix C and a more detailed description in the section "Intervention Design." All appendices are available at the end of this article as it appears in JPAM online. Go to the publisher's website and use the search engine to locate the article at http://onlinelibrary.wiley.com.

[7] Advising was provided by the Virginia College Advising Corps (VCAC), a chapter within the national College Advising Corps. The VCAC advisors completed additional training related to the military, military education benefit programs, military transitions, and the GI Bills prior to the pilot program. We describe the advising treatment in more detail in the section "Virginia College Advising Corps."

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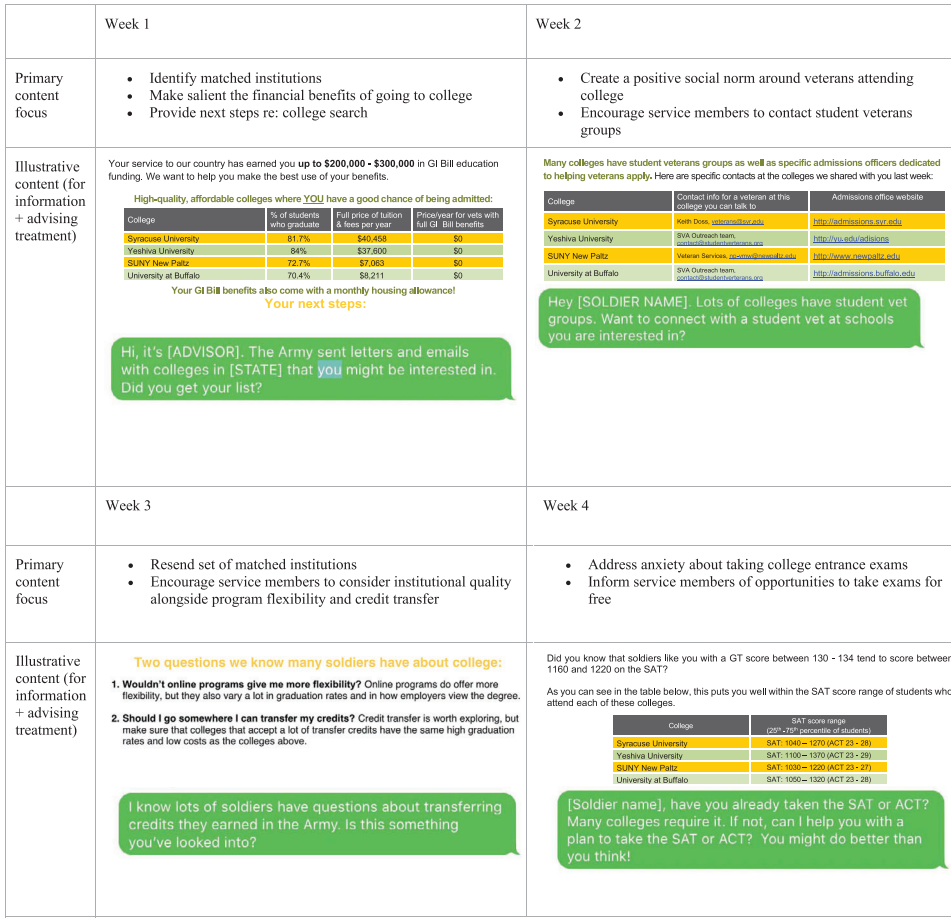

	Week 1	Week 2
Primary content focus	• Identify matched institutions • Make salient the financial benefits of going to college • Provide next steps re: college search	• Create a positive social norm around veterans attending college • Encourage service members to contact student veterans groups
Illustrative content (for information + advising treatment)	Your service to our country has earned you **up to $200,000 - $300,000** in GI Bill education funding. We want to help you make the best use of your benefits. **High-quality, affordable colleges where YOU have a good chance of being admitted:** College | % of students who graduate | Full price of tuition & fees per year | Price/year for vets with full GI Bill benefits Syracuse University | 81.7% | $40,458 | $0 Yeshiva University | 84% | $37,600 | $0 SUNY New Paltz | 72.7% | $7,063 | $0 University at Buffalo | 70.4% | $8,211 | $0 Your GI Bill benefits also come with a monthly housing allowance! **Your next steps:** Hi, it's [ADVISOR]. The Army sent letters and emails with colleges in [STATE] that you might be interested in. Did you get your list?	**Many colleges have student veterans groups as well as specific admissions officers dedicated to helping veterans apply.** Here are specific contacts at the colleges we shared with you last week: College | Contact info for a veteran at this college you can talk to | Admissions office website Syracuse University | Keith Doss, veterans@syr.edu | http://admissions.syr.edu Yeshiva University | SVA Outreach team, contact@studentveterans.org | http://yu.edu/admissions SUNY New Paltz | Veteran Services, newtitle@newpaltz.edu | http://www.newpaltz.edu University at Buffalo | SVA Outreach team, contact@studentveterans.org | http://admissions.buffalo.edu Hey [SOLDIER NAME]. Lots of colleges have student vet groups. Want to connect with a student vet at schools you are interested in?
	Week 3	Week 4
Primary content focus	• Resend set of matched institutions • Encourage service members to consider institutional quality alongside program flexibility and credit transfer	• Address anxiety about taking college entrance exams • Inform service members of opportunities to take exams for free
Illustrative content (for information + advising treatment)	**Two questions we know many soldiers have about college:** 1. **Wouldn't online programs give me more flexibility?** Online programs do offer more flexibility, but they also vary a lot in graduation rates and in how employers view the degree. 2. **Should I go somewhere I can transfer my credits?** Credit transfer is worth exploring, but make sure that colleges that accept a lot of transfer credits have the same high graduation rates and low costs as the colleges above. I know lots of soldiers have questions about transferring credits they earned in the Army. Is this something you've looked into?	Did you know that soldiers like you with a GT score between 130 - 134 tend to score between 1160 and 1220 on the SAT? As you can see in the table below, this puts you well within the SAT score range of students who attend each of these colleges. College | SAT score range (25th - 75th percentile of students) Syracuse University | SAT: 1040 – 1270 (ACT 23 - 28) Yeshiva University | SAT: 1100 – 1370 (ACT 23 - 29) SUNY New Paltz | SAT: 1030 – 1220 (ACT 23 - 27) University at Buffalo | SAT: 1050 – 1320 (ACT 23 - 28) [Soldier name], have you already taken the SAT or ACT? Many colleges require it. If not, can I help you with a plan to take the SAT or ACT? You might do better than you think!

FIGURE 1 Overview of intervention materials.

Notes: The SFL2 intervention consisted of postal and digital messaging, sent weekly over four weeks. The full set of intervention materials are available in the Appendix and a more detailed description of the intervention is on pages 8 through 10.
All appendices are available at the end of this article as it appears in JPAM online. Go to the publisher's website and use the search engine to locate the article at http://onlinelibrary.wiley.com.

13,173). Because our focus was on improving the quality and selectivity of service members' post-secondary enrollment as veterans, we began the intervention approximately 1 year, on average, before their actual separation date. We designed this gap between the intervention and separation to ensure service members had sufficient time to investigate their resources and support and to explore and apply to well-matched colleges, which may have annual or twice-annual application schedules. We restricted the sample to service members without a college degree and with an Armed Forces Qualification Test (AFQT) score (a subscore of the ASVAB) at or above the 65th percentile of a nationally normed sample.[8] The latter restriction was intended to focus on a subpopulation of veterans who appeared academically ready for college, with a significant scope for improvements in the quality of their college choices.

Despite high rates of mismatch in our focal sample and an intervention design incorporating strategies that have successfully overcome similar frictions in other populations, we find no overall impact

[8] As comparison, the 65th percentile of SAT score from a nationally representative sample of high school students is approximately 1090; as a result, service members in the sample were likely admissible to at least moderately selective institutions, despite historically low rates of veteran enrollment in these schools.

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of the intervention on whether service members enroll in college, on the quality of their college enrollment, on their persistence in college, or on their use of GI Bill education benefits. We have sufficient precision to rule out overall enrollment effects greater than 1.9 percentage points (3.4% relative to control mean). On the quality margin (i.e., enrollment in a college with a graduation rate over 30%), we can rule out effects larger than 0.98 percentage points (3.6% relative to control mean), significantly smaller than the effects estimated on similar margins in several recent studies with high-scoring traditional students. Effects on college persistence and GI Bill usage are similarly small.[9] We find no added benefit of offering service members proactive and personalized college advising support.

We can rule out that the null effects are because of a lack of exposure to the intervention materials: we were able to deliver text messages to 90% of both experimental groups; all nine text messages were delivered to over three quarters of both groups, indicating low opt out rates among treated service members (since recipients were given the option to opt out at any point). Among service members randomly assigned to the advising condition, two thirds of service members responded to at least one message. This rate of response is substantially higher than prior text-based advising interventions that have generated significant improvements in postsecondary participation: In those interventions, text response rates ranged from 19% to 48% (Castleman & Page, 2015, 2017; Page et al., 2020).[10] Our estimate of engagement is moreover conservative given that materials were also sent in paper and e-mail form, mediums for which we cannot track engagement given the specific delivery channels we used through the Army.

Qualitative analysis of the text message moreover suggest a meaningful contrast between the college and financial aid information and support the advising treatment received relative to the other experimental groups: nearly one-third of all service members assigned to the advising condition had substantive interactions with their advisor, in which the service member shared their specific interests in higher education, requested additional information as prompted by the automated messages, or asked unprompted questions. Advisors engaged with service members by asking questions to better understand their interest and needs, searching for and presenting a few college options, referring them to additional resources available, and inviting them to continue texting back with any questions. There were, however, a few notable limitations to the advising condition, including limited capacity of advisors to respond to service members' detailed questions about GI Bill funding and advisors' limited familiarity with technical training programs of interest to service members. These advising limitations may help explain the null effects for this experimental condition despite positive effects from prior similarly designed text-based college advising interventions. Specifically, service members may have been reluctant to actively explore higher-quality colleges and universities if they remained uncertain of how much GI Bill funding they could use at these institutions or if they did not believe these institutions offered trade-oriented postsecondary programs in which they were more interested.

Our intervention materials identified colleges and universities with objectively higher quality and affordability indicators in the communities where service members planned to live after separating from the Army, so the choices we identified did not require a trade-off between institutional quality and affordability or living in a desired geographic location (e.g., near family).[11] Given the low baseline institutional quality of enrollment among service members, and the fact that we leveraged (1) numerous communications channels, (2) multi-faceted evidence-based behavioral strategies that have

[9] We observe a small (roughly 1 percentage point) increase in degree attainment but view this as likely spurious given the number of hypothesis tests we conduct and the lack of impacts we observe on preceding outcomes (enrollment, enrollment quality, persistence, GI Bill benefits use) that would conceivably result in higher rates of attainment. These effects are concentrated among service members with substantial pre-intervention college enrollment, particularly prior enrollment at for-profit institutions. This result is somewhat surprising in the context of the motivation for the experiment, and further suggests that a lack of information is not the driving force behind the continued disproportionate enrollment of veterans in institutions with poor average outcomes.

[10] The response rates are similar to those (70%) in an advising intervention that resulted in reduced student loan borrowing, worsened course performance, and increased rates of student loan default (Barr et al., 2021).

[11] Over half of enrolled separating soldiers are enrolled within 100 miles of their planned location after separation. This is conservative lower bound given high rates of online enrollment. Separating soldiers could also use the provided website to search for schools in other locations.

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been shown to influence enrollment and college choice among traditional students, and (3) text-based advising, there was substantial opportunity and expectation to improve the quality of veterans' college enrollments. And yet, we find no impact on either overall enrollment or the quality of institutions veterans attended.

These results suggest that light touch strategies that have been effective at addressing similar challenges among traditional student populations, and which were adapted for military service members, are not sufficient to generate meaningful improvements in enrollment quality among veterans returning to civilian life. More specifically, information on academic match and college quality is unlikely to change service members' decisions, even when paired with behavioral strategies to address concerns about affordability or belonging. Providing additional light-touch, text-based advising from professional advisors was similarly ineffective in the context of our experiment; this may be attributable to advisors lacking sufficient training or experience providing college advising for military personnel.

Lower-intensity approaches to support veteran enrollment may particularly be insufficient to overcome inertia in enrollment (many service members who enrolled while in the military continued at the same institution after the intervention) or the strong inclination to enroll at for-profit institutions created by intensive recruiting (for-profits hire tens of thousands of recruiters hired for sales experience, not college advising) and by the misleading and inaccurate information recruiters provide about for-profit institution quality, cost, and credit transferability (Committee on Health, 2012). Earlier and/or more costly interventions, such as an Army-adapted version of evidence-based intensive college advising programs (delivered by more thoroughly-trained Army education counselors) or more assertive policies, such as those that impose tighter regulations on for-profit recruitment, constrain service members' postsecondary choices, or that tie financial aid receipt to prior institution performance, may be necessary to improve postsecondary education and employment outcomes among veterans.

BACKGROUND

Army partnership

We designed and executed our randomized controlled trial in conjunction with the U.S. Army Office of Economic and Manpower Analysis (OEMA) at West Point under the program "Soldier for Life, Student for Life," which builds on the Army's existing transition program brand of "Soldier For Life." OEMA secured program approvals from the Department of the Army, coordinated implementation with the Army's education and transition offices and military installations, distributed the program materials, created and monitored the program website, administered the program surveys, and collected and stored all program related data.[12]

Transition Assistance Program

Military service members who decide to leave active-duty are required to complete the Transition Assistance Program (TAP).[13] The purpose of TAP is to provide separating service members resources to successfully navigate their transition to civilian life, and offers a combination of job and career search, educational, and entrepreneurial guidance to service members.[14] Service members are

[12] OEMA merged and de-identified all program data prior to analysis. The research team accessed this data via secure methods on Army servers.

[13] This requirement was established under the National Defense Authorization Act of 1991 and the revisions of the Veterans Opportunity to Work Hire Heroes Act of 2011.

[14] The TAP is executed by each military service, coordinated by the Department of Defense, and includes contributions from other agencies including the Department of Education, Department of Labor (DOL), Veterans Affairs (VA), and the Small Business Administration.

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encouraged to begin the TAP no later than 12 months prior to their separation, and they are required to complete the TAP prior to separation. The TAP includes several required events (e.g., creation of a transition plan, VA benefits counseling, DOL employment transition session) as well as optional events. These optional events include a 2-day module in which service members can learn more about pursuing higher education options.[15] Take up of this module is very low: while over half of separating service members use their GI Bill benefits after separating, only 10% of service members attend the optional higher education module at the time of our intervention (GAO, 2017).

We designed our program materials to complement and not replace existing military programs and resources, including the TAP program and its optional higher-education module, by providing contact information for the service member's local education office and the transition assistance office in our intervention materials, and by coordinating our program launch and execution with the local education and transition counselors. Some aspects of the optional higher education module (including some focus on the use of graduation rates, employment information, and access to veteran supports in the college choice decision) are aligned with our intervention, so any shifts into the module among treated service members would be expected to reenforce the importance of college quality. While there is some descriptive research on the TAP (see Heflin et al., 2016, for a brief review), there is very little experimental or quasi-experimental evidence on the program's effectiveness.[16]

GI Bill education benefits

The 1944 Serviceman's Readjustment Act of 1944 and the Montgomery GI Bill of 1984, commonly known as the GI Bill or the Montgomery GI Bill (MGIB), established significant education benefits for eligible veterans. Under the MGIB, military service members must pay $100 into the program each month for 12 months, with the default set as opting-in to payments, to obtain non-taxable benefit payments directly from the VA to cover their living expenses and tuition and fees while enrolled in school. Previous research on the GI Bill documents significant labor market returns to the program (Angrist, 1993; Bound & Turner, 2002).

The Post 9/11 Veterans Educational Assistance Act of 2008 established the Post 9/11 GI Bill (PGIB). The PGIB offers more generous benefits in a slightly different structure. Service members can earn eligibility and use MGIB or PGIB benefits. Under the PGIB, service members are not required to pay into the program, and they gain eligibility based on the length of their honorable service. Average benefits levels are nearly twice as high under the PGIB, largely due to a locality-adjusted housing allowance and expanded tuition benefits authorized up to the most expensive in-state tuition rate for a public institution in the state. PGIB payments are delivered separately to the veteran (i.e., book stipend and housing allowance) and the institution (i.e., tuition and fees). Many institutions supplement the PGIB tuition payments with their own contributions if their tuition exceeds the state maximum under the Yellow Ribbon Program. Research has documented positive effects of the PGIB on enrollment and degree completion (Barr, 2015, 2019). However, recent work has indicated that the overall labor market effects of the benefit expansion were negative, with these effects driven by reduced work experience among veterans as well as low returns to the schools they attended (Barr et al., 2021). Particularly relevant for this paper, additional investments at for-profit institutions induced by the PGIB appeared to generate negative effects on subsequent earnings, underscoring the significant body of more descriptive evidence suggesting that these institutions generated poor outcomes.

Our program and materials were designed to support educational pathways using either version of the GI Bill, and the choice of which GI Bill Education Benefit to use was not a focus of the current study. Since the decision of which benefit to use, if eligible for both, can be complicated and unique

[15] For more details on the TAP, see: https://www.dodtap.mil/

[16] One exception is Li (2020), who found that the TAP increases veteran labor force participation.

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to each person, our materials and project advisors encouraged individuals to contact the education counselors at their military installation. That said, among veterans who have used GI Bill Education Benefits in recent years, over 90% of veterans have elected to use the PGIB. We also provided contact information for the VA and local education offices for individuals to determine their unique eligibility levels.

Virginia College Advising Corps

Service members randomly assigned to receive both personalized college information and advising received text-based outreach from advisors at the Virginia College Advising Corps (VCAC). VCAC is part of the national College Advising Corps, which typically hires recent college graduates and places them in under resourced high schools to increase college and financial aid application support for students. Our intervention materials introduced advisors as general college advisors, not specifically from the VCAC, to avoid potential confusion among service members about why they were being contacted by an advisor from Virginia. The VCAC advisors that took part in this pilot completed additional training related to the military, military education benefit programs, military transitions, and veteran education benefit programs (including the GI Bills) prior to the pilot program.

INTERVENTION DESIGN

Conceptual framework

Prior theoretical and empirical work from economics and social psychology suggests a number of factors that might contribute to veterans' disappointing postsecondary choices and outcomes. One factor contributing to observed undermatch may be veterans' lack of awareness of differences in college quality, the importance of college quality, or their potential admissibility to selective institutions. As with traditional-age populations, this lack of awareness may be due in part to insufficient access to college information or assistance within Army installations (as well as service members' high schools prior to the Army) and the fact that most Army installations are located outside major metropolitan areas, where selective institutions tend to be concentrated (Hoxby & Avery, 2012; Hoxby & Turner, 2013). Another factor may be that service members overestimate the net price they would pay to attend selective colleges and universities (Avery & Kane, 2004; Grodsky & Jones, 2007; Horn et al., 2003), or that service members are deterred by the perceived complexity of the process of applying for and receiving GI Bill benefits (Bettinger et al., 2012; Dynarski & Scott-Clayton, 2006).[17]

Social psychological and behavioral barriers may also impede military personnel from applying to or attending selective colleges or universities. Service members may face doubts or anxiety about whether they belong at colleges and universities where they perceive the student body to be predominantly younger and from different socioeconomic backgrounds (Walton & Cohen, 2007, 2011). Stereotype threat may lead service members to perform worse on college entrance exams if they internalize negative stereotypes about veterans' academic abilities (Aronson et al., 2002). Finally, service members may intend to apply to or attend selective institutions but miss important deadlines or processes because of limited attention or procrastination (Castleman & Page, 2015; Karlan et al., 2016).

[17] These studies document how complexities with the Free Application for Federal Student Aid (FAFSA) may deter traditional-age students from applying for aid or attending college.

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Institution matching algorithm

The primary purpose of the intervention was to provide service members with personalized information about well-matched colleges and universities in the communities they were returning to after the Army, to address the frictions and challenges we describe above.[18] We designed our matching algorithm to prioritize institutional quality and affordability, given prior work (cited above) indicating that (1) students are more likely to graduate from college and to realize greater labor market returns when they attend higher-quality institutions and (2) students are more likely to apply to and attend higher-quality institutions when they receive explicit, clear information that they can attend college tuition-free.

Specifically, we developed an algorithm to identify well-matched colleges (1) where a service member appeared likely to be admissible based on their academic performance and (2) that maximized a combination of quality (proxied for by institutional graduation rates) and affordability (i.e., price net of anticipated GI Bill funding). To generate the recommendations, our algorithm drew on service member attributes, including predicted SAT score (based on service members' General Technical [GT] score[19]) and the location where service members indicated they plan to reside after the Army, and institutional attributes, including institutional graduation rate and estimated price net of GI Bill funding. We identified service members as likely admissible at a given institution if their predicted SAT score was at or above the 25th percentile of the SAT scores of incoming freshmen.[20] We next restricted the set of potential matches to institutions with regional accreditations and with graduation rates above 30%.[21] We also eliminated primarily associate-granting institutions because of their generally low graduation rates and prior research demonstrating positive labor market returns to attending 4-year instead of 2-year institutions. We also eliminated institutions with specific focuses (e.g., theology).[22] Within this set of institutions, we generated a list of four institutions that included the top public institution and the top private institution within 50 miles of their location (zip code or city), as well as the top public and top private institutions in their intended state. The "top" option for public institutions was determined by sorting on graduation rate (the higher, the better); the top private institution was determined by giving equal weight to graduation rate and veteran net price (the lower, the better).[23] For the messaging content, we limited the number of recommended institutions to four to mitigate choice overload. We also included a link to a dedicated project website where service members could find the full list of matched institutions generated by the algorithm and/or generate a list of matched institutions for any location in the United States.

[18] When service members initiated their transition assistance program, they were asked to provide their intended location after separation by providing a zip code, city, or state. Sixty-three percent provided a zip code or city, 6% provided only a state, and the remainder provided no intended location. We used the most specific information the service member provided when generating the college recommendations. If a service member did not provide any location information when signing up for transition services, we used their home of record information on file. For service members with no home of record on file, we used the location of their current Army base. We find similar estimated treatment impacts for subgroups based on whether the service member provided their intended location or not (Appendix Table A1).

[19] While academic researchers are typically more familiar with Armed Forces Qualification Test (AFQT) scores, we use a service member's General Technical (GT) score since they are more widely used in the military (e.g., for selection into the military, bonus eligibility, and for classification into certain military career fields) and thus more well-known by service members. In practice, the two scores are very similar and both rely on similar underlying subject area tests from the broader Armed Services Vocational Aptitude Battery (ASVAB). The GT score combines three subtests (Arithmetic Reasoning, Paragraph Comprehension, and Word Knowledge) while AFQT combines four subtests (the same three above plus Mathematics Knowledge).

[20] To generate service members' predicted SAT score, we used data from the College Board to merge in SAT scores for applicable service members in historical cohorts. We then regressed SAT scores on a quartic model of GT scores to develop a GT–SAT concordance table. For example, a GT score of 117 (roughly the median of service members eligible for our sample) corresponds to a predicted SAT score of 1000 on the 1600 scale.

[21] We intended to set a higher graduation rate threshold for recommended colleges, such as 40% or 50%, which would be closer to the median graduation rate at the time of our intervention. However, these higher thresholds were not inclusive of several state flagship universities and resulted in fewer than four recommended colleges for a large portion of the sample.

[22] While we do not explicitly exclude for-profit institutions from this list of colleges, these restrictions resulted in a very small number of for-profits being included (3 out of 800 institutions).

[23] Private veteran net price was scaled relative to the maximum veteran net price in the sample before weighting.

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Outreach materials

We sent each treated service member a combination of postal mail, emails, and text messages. All treated individuals received weekly postal mailers and emails for 1 month (totaling four letters and four emails) and twice-weekly text messages (totaling nine messages, which included one introductory text). Each week's postal and email content were identical, just delivered through different channels. The texts for the information-only treatment arm prompted service members to give a closed-ended response (e.g., "Yes" or "No") to receive information about matched institutions. The texts for the information+advising treatment arm were conversational and framed as coming from a particular advisor. They prompted service members with open-ended questions and explicitly invited service members to engage directly with their advisor. We include a full set of intervention materials in Appendix C.

As we reference above, we designed the materials to leverage a combination of evidence-based behavioral science strategies to strengthen service member's engagement with and responsiveness to the materials. The first week's letter, email, and texts had four main objectives. First, they reinforced the financial benefits of obtaining a college degree with a concrete estimate of the lifetime earnings premium service members could expect from earning a degree. Second, they attempted to create an endowment effect by defining the amount of GI Bill service members had earned through their military service. Third, they delivered the list of matched institutions. We included the institution name, graduation rate, sticker price (i.e., tuition and fees), and price net of GI Bill funding. We included graduation rates to provide an easily understood measure of college quality.[24] The price net of GI Bill funding was often $0, providing a clear indication that the investment in education would not incur a direct financial investment by service members. We included the sticker price to provide a concrete reference point for the magnitude of the price reduction service members could realize by using their GI Bill funding. Finally, the first set of communications provided concrete next steps, including encouragement to search for additional well-matched schools on the dedicated project website (where they could view the same college-level information for the full set of recommended colleges for any location in the U.S.) and to attend the higher education transition module at their installation as part of their transition process.

The second week's materials aimed to create a positive social norm of veterans attending college by providing specific veterans' group contacts at each of the four institutions we identified in the first week's materials. We also normed that the broad set of skills service members developed through the Army could support their success in college, and encouraged service members to contact student veterans' groups as a next step.

The third week's materials repeated the set of matched institutions with graduation rate and net price from the first week and addressed common questions service members often have about postsecondary educational options, specifically whether they should prioritize in their search flexible online options and institutions that are more flexible in their acceptance of transfer credits.

The fourth and final week's materials aimed to address potential anxiety among service members about whether they would score sufficiently high on college entrance exams by providing each service member an estimate of their SAT score range based on a concordance we calculated between service members' GT scores and SAT scores, and how these ranges map to the interquartile range of SAT scores at the matched list of institutions. We also informed service members that they may be able to take the SAT for free at their installation's education office.[25]

[24] Graduation rates correlate strongly with measures of instructional expenditures, post-college earnings and loan repayment, and even estimates of value-added (Rothwell & Kulkarni, 2015).

[25] Our data (described in more detail below) included baseline SAT scores for service members who took the exam prior to joining the Army. As we show in Table 1, mean SAT scores are statistically equivalent between groups. Our data did not include SAT or ACT college entrance exam scores taken during or after service members' enlistment, so we are not able to estimate impacts of the intervention on college entrance exam taking or performance.

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The postal letters and emails were identical in content for both treatment groups except that the information and advising letters and emails included encouragement to contact an advisor as a next step, along with the relevant phone and email contact information.

EXPERIMENTAL DESIGN

Our experimental sample is comprised of service members who entered the TAP to begin their separation from the Army and prepare for their transition back to civilian life. We defined service members as eligible for the intervention if they had not already earned a college degree and had an Armed Forces Qualification Test (AFQT) score (a subscore of the ASVAB) at or above the 65th percentile. The latter restriction was intended to focus the sample on a subpopulation of veterans who had significant scope for improvements in the quality of their college choices. The SAT 65th percentile score for a nationally normed sample is approximately 1090; as a result, service members in the sample were likely admissible at moderately or more selective institutions with high graduation rates and instructional spending per student, despite their tendency to enroll at these institutions at low rates.

We conducted the intervention at six of the largest Army installations across the United States: Forts Bragg (near Fayetteville, North Carolina), Campbell (near Clarksville, Tennessee), Carson (near Colorado Springs, Colorado), Hood (near Killeen, Texas), Stewart, and Hunter Army Airfield (both near Savannah, Georgia). Because Army personnel enter the TAP on a rolling basis over the calendar year, we conducted our randomization on roughly a monthly basis from March 2016 to November 2017, for a total of 17 waves. In each wave, we randomly assigned service members to one of three experimental groups (control, information-only, information+advising) in equal proportion. We performed our randomization within wave by installation by GT score quartile blocks.[26] We chose to randomize within GT score quartiles because we expected the counterfactual enrollment and graduation outcomes to differ meaningfully by service member aptitude, with higher aptitude individuals achieving better outcomes. Indeed, among our control group, service members in the highest GT quartile are 21% more likely to enroll and 29% more likely to earn a degree after the intervention, compared with service members in the lowest GT quartile.[27]

Our resulting experimental sample size is $n = 13{,}173$. The control group consists of 4,381 service members; the information-only treatment group consists of 4,389 service members; and the information+advising treatment group consists of 4,403 service members.

DATA

Our data come from four primary sources. Service member–level administrative data from the U.S. Army provided baseline data for the experimental sample including service members' race/ethnicity, gender, number of dependents, GT score, monthly pay, years of service, rank, and the number of months service members spent in hostile environments. Text interaction data from the Signal Vine platform provided information on the share of service members that were sent and responded to the intervention text messages. The National Student Clearinghouse Research Center (2017, 2019) provided student by term-level college enrollment data, with coverage of institutions that account for

[26] We used four GT bins that correspond to the quartiles of GT scores within AFQT CAT I and II (based on a national sample from active service members in September 2015): (1) less than 113; (2) 113 to 116; (3) 117 to 122; and (4) 123 or higher. Interacted with site by wave, there are 408 randomization blocks.

[27] We test whether our experimental impacts differ across the four GT quartiles by estimating separate regression models for each of the four bins. We find no consistent patterns indicating that the intervention had differential impacts by service member aptitude.

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TABLE 1 Baseline equivalence and summary statistics for the control group.

	Info only		Plus advising		Control mean	Coef Test *P*-value
	(1)	(2)	(3)	(4)	(5)	(6)
White	0.0038	(0.0092)	0.0076	(0.0092)	0.743	0.678
Black	0.0021	(0.0066)	0.0056	(0.0066)	0.105	0.599
Hispanic	−0.0068	(0.0062)	−0.0189***	(0.0062)	0.102	0.0517
Other race	0.0009	(0.0047)	0.0057	(0.0047)	0.0493	0.305
Female	0.0085	(0.0059)	0.0097	(0.0059)	0.0797	0.852
Ever married	0.0069	(0.0106)	0.0103	(0.0106)	0.541	0.743
Has dependents	0.0187*	(0.0105)	0.0179*	(0.0105)	0.569	0.941
GT score	−0.1089*	(0.0583)	0.0058	(0.0582)	117.5	0.0487
Base monthly pay	4.4284	(15.5270)	−9.5507	(15.5208)	2580	0.368
Years of service	0.0041	(0.1081)	−0.0744	(0.1080)	5.436	0.467
Hostile fire pay (months)	0.1030	(0.2519)	−0.2909	(0.2518)	7.700	0.118
TA credits	0.2528	(0.4179)	−0.1365	(0.4174)	8.312	0.351
SAT score	−3.6064	(5.9362)	−3.4164	(6.0010)	1027	0.975
E01 to E04	0.0057	(0.0101)	0.0210**	(0.0101)	0.645	0.131
E05 to E06	−0.0014	(0.0096)	−0.0148	(0.0096)	0.292	0.165
E07 to E09	−0.0043	(0.0051)	−0.0062	(0.0051)	0.0635	0.706
Prior days enr, grad rate < 30%	−2.3082	(5.5228)	−4.4267	(5.5170)	165.4	0.701
Prior days enr, grad rate 30–50%	−3.4813	(2.8551)	−1.0793	(2.8521)	42.70	0.400
Prior days enr, grad rate > 50%	−2.7776	(3.6617)	−6.5761*	(3.6579)	56.27	0.299
N	4,389		4,403		4,381	

Notes: Each row corresponds to a separate regression of the baseline variable on the two indicators for treatment assignment (info only, plus advising) and randomization block fixed effects (base x GT bin x intervention wave). Columns 1 and 3 display the coefficient estimates for the treatment assignment indicators, and columns 2 and 4 display the standard errors. Column 6 displays the "Coef Test P-value" is the *p*-value from the test that the coefficients for the Info only and Plus advising indicators are equal. All baseline variables are measured immediately prior to the relevant intervention wave launch. Tuition Assistance (TA) credits refer to college credits the soldier earned through the Army's Tuition Assistance program. SAT scores are linked using data from the College Board. E01 through E09 categories refer to rank, with E01 being the lowest. The "prior days enrolled" measures are constructed using National Student Clearinghouse matches in the same manner as the weighted days enrolled outcome measures. The sample size for each regression is $n = 13{,}173$ with the exception of Base monthly pay ($n = 13{,}067$), Hostile fire pay ($n = 13{,}067$), and SAT score ($n = 2{,}134$).

96.1% of college enrollments in the country.[28] Finally, we incorporate administrative data from the Department of Veteran's Affairs on monthly payments made as part of the Post 9/11 GI Bill.[29]

SAMPLE AND BASELINE EQUIVALENCE

In Table 1, we present descriptive statistics on our sample and report results from models in which we regress student-level baseline characteristics on the treatment indicator and wave by installation by GT bin fixed effects. The sample is primarily male (92%) and majority White (74%). Just over half of the sample (54%) had ever been married and 56% reported dependents. The mean service member in

[28] NSC data also include most online enrollments. In Fall 2017, the coverage rate for "Multi-state or Primarily Online Institutions (POIs)" was 84.2%. Source: https://nscresearchcenter.org/workingwithourdata/

[29] Payment data (as opposed to months of benefits used) for the Montgomery GI Bill was not available for analysis. As mentioned above, the vast majority of recently separated veterans use PGIB benefits instead of MGIB.

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the sample had served for just over 5 years and received just over $2,500 in base monthly pay from the Army. Across 19 baseline measures we only find two significant differences at the 0.05 level between the treatment and control group. Treated service members are slightly less likely to be Hispanic and slightly more likely to have lower rank within the Army. We include covariates in our models both to account for this slight imbalance and to improve our precision.[30]

Both through postsecondary education individuals pursued before joining the Army and through the Army's Tuition Assistance (TA) program, a meaningful share of service members participate in postsecondary education and training prior to separating from the Army. On average, service members in our sample spent just over a year (377 days) enrolled at a higher education institution prior to the start of the intervention, though none had completed a bachelor's degree prior to the intervention. While this enrollment could have occurred prior to the service member's enlistment, we do observe meaningful engagement in higher education while in TA: on average, service members in our sample accumulated an average of 8.3 credits of education and training through TA. Prior and concurrent postsecondary education and training are balanced across our experimental conditions.

College enrollment is fairly high among this population of service members: Among our control group, 56% enrolled after the intervention was complete. Among those who did enroll, 41% enrolled at a public 4-year; 18% enrolled at a private nonprofit 4-year; 47% enrolled at a community college; and 18% enrolled at a for-profit. Only one third of the control group service members who enrolled did so at one of the colleges that met the criteria for our recommendation lists.

EMPIRICAL STRATEGY

We estimate the impact of providing service members personalized information and advising about college options on a variety of college enrollment, enrollment quality, persistence, and completion measures. Our primary specification is:

$$Outcome_i = \beta_0 + \beta_1 Treatment_i + \beta_2 X_i + RandomizationBlockFE_i + \epsilon_i \quad (1)$$

where $Outcome_i$ is generally a college enrollment outcome for individual i and X_i includes the baseline characteristics from Table 1.[31] We include randomization blocks fixed effects to account for the level at which we randomized service members to the intervention. We estimate two versions of this model, one which pools both treatment arms into an "any treatment" indicator and another which includes separate indicators for each of the two treatment arms (information-only, information+advising). These coefficients represent the intention to treat (ITT) estimate of being assigned to a treatment condition.[32]

Our key outcomes of interest are measures of college enrollment and enrollment quality, specifically indicators for whether service members attended one of the four colleges from their personalized recommendation list; any 4-year institutions with graduation rates above 30%; or any for-profit institutions.[33] We also present estimated effects on weighted days of enrollment, GI Bill usage, and degree attainment. We measure and report our outcomes separately based on whether they occurred after the intervention, after the intervention and before service members separated from the Army, or after service members separated.[34]

[30] We also test baseline equivalence for assignment to any treatment versus control and see similar patterns.

[31] For the three baseline characteristics with missingness—Base Monthly Pay, Hostile fire pay, and SAT score—we set the values of these characteristics to zero and include missing indicators. We have SAT scores (measured prior to the intervention) for 16% of the experimental sample. Base monthly pay and hostile fire pay are missing for fewer than 1% of observations.

[32] We also estimated logistic regression models and found very similar results.

[33] Specifically, we use the graduation rate in 150% of formal time at 4-year institutions, retrieved from the College Scorecard.

[34] One concern may be that the intervention affected the service member's choice to stay in the Army by re-enlisting or attempting to separate sooner. In Appendix Table A3 we test whether the intervention impacted separation timing and find no evidence to support this hypothesis.

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TABLE 2 Text message interaction statistics.

	Info only (1)	Plus advising (2)
Received any texts	89.4%	90.0%
Received all nine texts	80.3%	74.6%
Any response (conditional on receipt)	17.2%	65.7%
Conditional on any response		
Total number of text responses	3.1	4.2
Total length of text responses (characters)	23.5	157.8
Responded to specific intervention text (conditional on receipt)		
1	1.1%	13.1%
2	18.4%	44.3%
3	1.1%	36.2%
4	3.3%	20.7%
5	3.0%	13.5%
6	0.3%	18.0%
7	0.3%	13.3%
8	0.3%	15.3%
9	0.2%	17.0%
N	4,389	4,403

Notes: Calculated using text interaction data. The ∼10% of soldiers who did not receive any texts were due to invalid cell phone numbers. Soldiers who received all nine texts are those who had a valid cell phone number and did not opt-out of the text messages at some point during the intervention. Response to a particular intervention text is measured by whether the soldier texted back before the subsequent program message was sent.

RESULTS

Intervention engagement

In Table 2 we present text message engagement statistics separately for the information-only and information+advising treatment arms. We successfully delivered at least one intervention text to approximately 90% of both groups. All nine campaign texts were delivered to the substantial majority of both groups (80% for the information-only treatment; 75% for the information+advising treatment), indicating that opt-out rates were relatively low among treated service members. Response rates were much higher in the information+advising group (as intended). Two thirds of service members in the information+advising group responded to at least one text, compared with only 17% of service members in the information-only group. Conditional on ever responding, service members in both treatment arms sent relatively few responses (three on average for information-only service members; four for information+advising service members), with responses tending to be quite short. The mean total number of characters in service members' responses in the information-only treatment was 24, which largely consisted of service members replying to the keyword prompts contained in the information-only messages. In the information+advising treatment the mean total number of characters was 158; across an average of four responses these were relatively short responses from service members either to the scheduled messages or advisors' queries. These statistics provide conservative estimates of engagement given that we also sent all intervention materials in paper and e-mail form,

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mediums for which we cannot track engagement at the individual level, but which only would have expanded engagement.[35]

We supplement these engagement statistics with a qualitative analysis of the interactions between service members and advisors in the information+advising treatment arm. We describe our approach to this qualitative analysis and the associated findings in more detail in Appendix B. At a high level, our examination of service member and advisor interactions indicate that a sizeable share of service members (29.4%) used the text message channel to engage in substantive interactions with a college advisor. In these interactions, service members requested additional information as prompted by program messages; expressed interest in discussing postsecondary options; or posed a specific question about the college application process.

Among service members requesting more information, there was particular interest in veteran groups or admission contacts (54.3%) and the GI Bill and federal financial aid (54.3%). Among service members with specific questions about the enrollment or financial aid process, over one third of questions (37.5%) were about GI Bill benefits or other Army-specific topics. Advisors were generally responsive to service members' requests for additional information or to their questions, e.g., by responding with probing questions to better understand service members' interests and needs or by presenting additional college options. However, we note three limitations we observed in advisor responses. First, advisors seemed less familiar with vocational and technical training programs some veterans expressed interest in; this is likely reflective of the advisors' primary experience focusing on 2- and 4-year academic program offerings with traditional high school students. Second, for military-specific questions (e.g., about their specific GI Bill eligibility), the advisors needed to refer the service members to Veterans Affairs or SFL-TAP in all cases. This was in line with the Army-specific training the advisors received prior to the start of the intervention: Due to the complexity of determining a specific individual's precise GI Bill eligibility, advisors were trained to emphasize that service members should contact Veterans Affairs with their specific questions. Third, some service members (roughly 20% of the full qualitative analysis sample) indicated they already had concrete plans for higher education and in some cases had already enrolled, even though on average the intervention began approximately a year prior to when service members separated from the Army. Advising about college options may have been less useful to this subset of service members.

An open question is whether the personalized college recommendations were relevant to a veteran's decisions. In particular, it is possible that service members moved somewhere else geographically than where they indicated they planned to move when they began the TAP. While we are unable to assess this directly,[36] a closer investigation of the text engagement data suggests that the recommendations were relevant. In both groups the highest response rates were to the second text message, which included the personalized college recommendations.[37] Furthermore, our qualitative review of these messages provides little support for the notion that the recommended colleges were irrelevant to a veteran's decision or that the assigned location was incorrect: Only 2.7% of service members in the advising+information group indicated that the location of the recommended colleges was either inaccurate or no longer relevant to their situation. While it is possible that some veterans received information inconsistent with their preferred location after separation, the project website made it easy for service members to identify well-matched institutions in other geographic areas.

[35] We used home addresses from personnel data, and we used official Army email addresses, which service members are required to maintain, or civilian email addresses, if provided by the service member when they enrolled in the TAP.

[36] We do not know of research investigating the consistency of service members' intended geographic residence as of the start of TAP and when they ultimately transition—on average a year later—so it is difficult to assess this possibility directly.

[37] In the information+advising group response rates were also relatively high for the third message, which inquired as to whether the service member attended the higher education transition module. We are unable to observe attendance at this higher education module.

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TABLE 3 Impact on enrollment and enrollment quality.

Panel A: Overall treatment impacts

	Any enrollment	Enrolled at inst with grad rate > 30%	Enrolled at for-profit
	(1)	(2)	(3)
Any treatment	0.0018	−0.0055	−0.0008
	(0.0088)	(0.0078)	(0.0054)
N	13,173	13,173	13,173
R-squared	0.1171	0.1191	0.0858
Control mean	0.559	0.273	0.101

Panel B: Impacts by experimental variation

	Any enrollment	Enrolled at inst with grad rate > 30%	Enrolled at for-profit
	(1)	(2)	(3)
Info only	0.0032	−0.0061	−0.0026
	(0.0101)	(0.0091)	(0.0063)
Plus advising	0.0004	−0.0049	0.0010
	(0.0101)	(0.0090)	(0.0063)
N	13,173	13,173	13,173
R-squared	0.1171	0.1191	0.0858
Control mean	0.559	0.273	0.101
Coef Test *P*-value	0.783	0.902	0.569

Notes: Within each panel, each column corresponds to a separate regression of the outcome variable indicated in th e column header on indicator(s) for treatment assignment, baseline variables shown in Table 1, and randomization block fixed effects (base x GT bin x intervention wave). In Panel B, the "Coef Test *P*-value" is the *p*-value from the test that the coefficients for the Info only and Plus advising indicators are equal.

Impacts on college enrollment and enrollment quality

In Table 3 we present impact estimates of the intervention on whether service members enrolled in college and on the quality of their college enrollment. We do not observe any impact of the intervention on overall enrollment: 56% of both treated and control service members enrolled in college at some point following the intervention. Based on our confidence intervals we can rule out treatment impacts larger than 1.9 percentage points, which would reflect a 3.4% when compared to the control mean. Despite the relatively low baseline enrollment quality of service members, we also observe no impacts of the intervention on whether service members attended an institution with a graduation rate greater than 30% (the threshold we used to include colleges on the recommended list; column 2) or on whether service members enrolled at for-profit institutions (column 3). As we show in the bottom panel of Table 3, we do not observe differential effects of the intervention based on whether service members were assigned to the information-only or information+advising treatment arms. We similarly find no enrollment impacts on enrollment at 2-year or 4-year institutions; public or private nonprofit institutions; institutions with graduation rates above 50%; or at one of the four colleges listed on a service member's personalized list (Appendix Table A2).

We can rule out effects larger than a percentage point on the institutional quality margins in columns 2 and 3 (3.6% and 10.2% relative to the control mean, respectively), significantly smaller than the magnitude of effects estimated in most similar studies among high-scoring traditional students. For example, Hoxby and Turner (2013) estimated a 5.3 percentage point (18.5%) increase in enrollment at well-matched institutions. Two other studies that evaluated the effect of an offer of virtual advising,

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TABLE 4 Impacts on college persistence.

	Days enrolled	Days enrolled at inst > 30% grad rate	Days enrolled at For-profit 4-year	Any 9/11 GIBill benefits used	Months of 9/11 GI Bill benefits used
	(1)	(2)	(3)	(4)	(5)
Any treatment	−1.8366	−3.2301	0.6821	0.0083	0.1154
	(4.5805)	(3.6186)	(1.4191)	(0.0091)	(0.1540)
N	13,173	13,173	13,173	13,173	13,173
R-squared	0.1223	0.1115	0.0628	0.0727	0.0981
Control mean	185.2	95.57	15.58	0.488	6.472

Notes: Each column corresponds to a separate regression of the outcome variable indicated in the column header on indicator for treatment assignment, baseline variables shown in Table 1, and randomization block fixed effects (base x GT bin x intervention wave). The "Days enrolled" outcome is the number of days from the start to the end of an enrollment period (not the number of instructional days), and is weighted by enrollment status (e.g., 0.5*days enrolled for half-time enrollment).

one component of our multifaceted intervention, found positive effects of 2.7 percentage points (5%) and 1.3 percentage points (2.5%) on enrollment in high graduation rate colleges, respectively (Gurantz et al., 2020; Sullivan et al., 2021). Concrete indications of tuition-free enrollment (also a component of our intervention) generated very large enrollment effects (15 percentage points, about 125%) at the University of Michigan, in a study that provided targeted encouragement and a more explicit commitment from a single institution if admitted (Dynarski et al., 2021).

The intervention was primarily intended to influence the post-separation behavior of service members. In Appendix Table A3, we demonstrate that assignment to treatment had no effect on a service members timing of separation. In Appendix Table A4, we decompose the same enrollment outcomes from Table 3 by whether they occurred after the intervention and *before* service members separate from the Army or after the intervention and *after* service members separate from the Army. We see no significant impacts on enrollment either before or after the point of separation.

In Table 4 we show impacts of the intervention on a variety of measures of college persistence, pooling across treatment arms. In column 1 we present impacts on the total number of days following the intervention that a service member was enrolled at any higher education institution, weighted by enrollment intensity.[38] Mirroring Table 3, columns 2 and 3 of Table 4 present impacts on days enrolled at institutions with at least a 30% graduation rate and days enrolled at for-profit institutions, respectively. In columns 4 and 5 we estimate impacts of the intervention on military-specific measures of college participation: whether service members used any post-9/11 GI Bill benefits towards a college education (column 4) and the number of GI Bill benefit months used by service members (column 5). Across measures we find no impact of the intervention. Nearly half (48.9%) of the control group had used GI Bill benefits and we can rule out treatment impacts on GI Bill use greater than 2.6 percentage points, a 5% increase when compared to the mean. As we show in Appendix Table A3, we also do not observe significant impacts of the intervention on the number of days enrolled when we decompose this outcome as occurring before or after service members separated from the Army. In Appendix Table A5 we show that the impacts of the intervention on our college persistence measures do not vary by treatment arm.

In Table 5 we present impacts of the intervention on three measures of degree attainment, pooling across treatment arms: whether service members earned any degree within our observation window (columns 1 through 3); whether service members graduated from an institution with a graduation rate of at least 30% (columns 4 through 6); and whether service members graduated from a for-profit institution (columns 7 through 9). Within each outcome the first column presents estimates of impacts

[38] The "Days enrolled" outcome is the number of days from the start to the end of an enrollment period (not the number of instructional days), and is weighted by enrollment status (e.g., 0.5*days enrolled for half-time enrollment).

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TABLE 5 Impacts on degree attainment.

	Earned degree			Degree from inst with grad rate >30%			Degree from for-profit 4-year		
	After intervention	Before separation	After separation	After intervention	After int, before sep	After separation	After intervention	Before separation	After separation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Any treatment	0.0096*	0.0081**	0.0035	0.0059	0.0021	0.0046	0.0064**	0.0045**	0.0012
	(0.0058)	(0.0037)	(0.0051)	(0.0039)	(0.0023)	(0.0034)	(0.0028)	(0.0022)	(0.0019)
N	13173	13173	13173	13173	13173	13173	13173	13173	13173
R-squared	0.1441	0.1259	0.0854	0.132	0.1039	0.0819	0.0974	0.0836	0.05
Control mean	0.121	0.0415	0.084	0.0477	0.0155	0.0336	0.0203	0.0126	0.00959

Notes: Each column corresponds to a separate regression of the outcome variable indicated in the column header on indicator for treatment assignment, baseline variables shown in Table 1, and randomization block fixed effects (base x GT bin x intervention wave). All outcomes are measured after the intervention. If a soldier had not yet separated as of March 2020 (the most recent NSC data to which we have access), then their "After separation" outcomes are recorded as zero.

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occurring after the intervention, while the second two columns decompose each outcome into whether it occurred before or after the service member separated from the Army. As we show in column 1, we estimate a marginally significant 0.96 percentage point increase in the probability of service members earning any degree, which represents an 8% increase relative to the control group. This slight increase in degree attainment is entirely driven by service members who earned a degree after the intervention but before their separation from the Army. We find no impact of the intervention on whether service members graduated from an institution with a graduation rate of 30% or higher, but do estimate a significant 0.6 percentage point increase in degree attainment from a for-profit institution.[39,40]

We view the impacts on degree attainment as likely spurious given the number of hypothesis tests we conduct and the lack of impacts we observe on preceding outcomes (enrollment, enrollment quality, persistence, GI Bill benefits use) that would conceivably result in higher rates of degree attainment. One possibility is that the intervention motivated service members to complete a credential or degree towards which they had already made substantial progress. We do find that degree impacts are concentrated among service members with substantial pre-intervention enrollment histories, and in particular for those with pre-intervention enrollment at for-profit institutions (Appendix Table A8). It therefore seems likely that any modest overall degree attainment impacts we observe, if real, are a function of the intervention nudging service members who were already close to a credential to either take the additional courses they needed through Tuition Assistance or gain credit for additional military experience they had accumulated—behavior we are unable to observe in our data. The fact that these additional investments appear to be coming through for-profit institutions is somewhat surprising in the context of the motivation for the experiment, and further suggests that a lack of information is not the driving force behind the continued disproportionate enrollment of veterans in institutions with poor average outcomes.

We also explore whether enrollment, persistence, and graduation impacts varied based on differences in the personalized list of recommended colleges in the intervention materials. Because service members in our sample have a wide range of intended locations after the Army (representing all 50 states, the District of Columbia, Puerto Rico, and Guam), the average quality of the four recommended colleges varies even holding constant service member ability. For example, for a service member with a GT score of 116 (the median of our sample), the average graduation rate of the college lists ranges from 36.9% to 74.3% (mean 55.5%, standard deviation 6.5) based on their intended location after the Army. To test for heterogeneity of impacts based on this dimension, we create quartiles of the average graduation rates of recommended colleges within each GT score. We then estimate the impacts separately by quartiles. As we show in Appendix Table A9, there are no heterogeneous impacts by quality of recommended colleges.[41]

DISCUSSION

We implemented a multi-faceted randomized control trial to evaluate the effects of information and advising services on the postsecondary choices and attainment outcomes of transitioning Army service members. To our knowledge, the intervention is the first large field experiment to study the impact of information and assistance intended to alter the quality of college choices of and improve enrollment quality among military service-members. Our sample of transitioning service members had

[39] As we show in Appendix Table A6, the impacts of the intervention on attainment are very similar across the information-only and information+advising treatments.

[40] As we show in Appendix Table A7, these increase in degree attainment show up 18 months after the intervention.

[41] We also repeat this analysis using variation in the share of veterans enrolled at the recommended colleges (a college-level variable from the College Scorecard), recognizing that service members may be more familiar or comfortable with a college with a larger share of veterans already enrolled. However, there is significantly less variation in this college-level attribute: the average share of veterans of the college lists ranges from 0.1% to 3.2%, with a mean of 0.5% and standard deviation of 0.3. In Appendix Table A10, we use a similar approach as with the graduation rates to show that there are no differential impacts on this dimension either.

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high observed rates of mismatch, suggesting significant scope for improvements in enrollment quality. Further, our intervention conveyed to service members that they would face a net price of $0 at most colleges and universities *without* having to complete additional financial aid applications; leveraged multiple communications strategies (i.e., postal mail, email, and text messages); evidence-based behavioral approaches (i.e., personalized information, social norming, simplification, and specific action steps); and professional advising resources to increase service members' pursuit of college admission, attendance, and completion at higher quality institutions. Our randomized design and large samples allow us to rule out any economically meaningful effects of the intervention on overall enrollment, institutional quality, or persistence (days enrolled or benefits used). Program engagement data enable us to rule out a lack of material receipt or engagement as an explanation.

Our results provide new evidence on the challenges associated with affecting the postsecondary education decisions of military service members. While our experiment was not designed to explain the absence of effects, we can provide some conjecture for our results. One possibility is that veterans feel out of place at the institutions we recommended. The provision of explicit connections to student veteran groups at the recommended institutions was intended to address this concern, but perhaps it was not enough to overcome separating service members' sense that few of their peers were attending these types of schools. Another possibility is that veterans are prioritizing some other institutional attribute. Older and younger students both rank academic reputation as the most important factor in their college choice, but older students are significantly more likely to rank convenience (e.g., availability of evening and weekend courses) as important in their college choice (Bryant, 2012). While veterans and other non-traditional students may place a higher value on social fit and flexibility in their college choice decisions, this may reflect an overweighting of immediate concerns versus a more comprehensive assessment of which institutions offer the highest probability of longer-term success. Indeed, for-profits heavily market flexibility as one of the reasons people should attend these institutions, yet students who attend for-profits and more flexible open access institutions still report regretting their institutional choices at high rates.

Another potential explanation for our null results is that service members simply default to continue their education at the institution, or the type of institution, they enrolled at during the Army. As we discussed above, many service members take advantage of the Tuition Assistance program to earn college credits while they are on active duty. Among service members who were enrolled in the two years prior to the intervention and who enrolled after the intervention (encompassing roughly 20% of our sample), nearly half (47.9%) enrolled at the same institution after the intervention as they had attended prior; over one third (36.9%) of these re-enrollments were at for-profit institutions. In this respect, interventions and programs designed to improve service members' educational decisions might need to begin earlier in their military tenure. That said, we see no difference in the effects of the intervention when we split the sample based on the quantity or quality of pre-intervention enrollment.

Specific to the information+advising experimental arm, one potential reason for the lack of efficacy of personalized advising is the unique institutional context. The VCAC advisors were very invested in and dedicated to the project, and as we describe earlier, staff from the Army continuing education office provided a thorough training to the advisors prior to the start of the intervention. That being said, service members differ significantly from the traditionally-aged college students the VCAC advisors are more experienced in working with, in terms of their interests and needs. In addition, some service members had already made concrete education choices before the intervention began. From a broader design perspective, this pattern of interactions highlights the importance of engaging advisors with sufficient knowledge and experience to address the context-specific questions that may be unique to certain non-traditional student populations like military service members and doing so at the right time.

In contrast, we view geographic mismatch as a relatively unlikely explanation given up to date information on intended geographic residence post-separation and easy access to the project website that allowed treated service members to generate a full list of recommendations for any location in the United States. While we do not observe service members' post-separation addresses, we can compare

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their intended location with the location of the institution they enroll in after separation. We find that half of all post-separation service members who enroll post-separation do so at an institution within 100 miles of their intended city or zip code. This is likely a lower bound on the quality of the geographic "match" of our recommended colleges, since individuals may have returned "home," but enrolled online. Also, less than 3% of service members in our qualitative text review sample indicated that the location of the recommendations was either inaccurate or no longer relevant. We also view as less plausible the possibility that veterans applied to and were not admitted to the institutions we identified or to similar quality institutions. By design, veterans should have been at least above the 25th percentile of the SAT score distribution at the institutions we featured, and many institutions give preference to military experience in their admissions decisions. We also view as less plausible that service members are already making optimal college choices from the perspective of maximizing their labor market potential. While there is relatively little research on the returns to college quality specifically for service members, Barr et al. (2022) showed that the returns for veterans attending for-profit institutions are actually negative, while veterans who attend public institutions do accrue some labor market benefits. Though we designed our intervention to better inform service members about the importance of college quality, the lack of enrollment impacts suggests our materials and advising were either not effective in changing service members' views on the returns to quality, or not effective at shifting their priorities when selecting a college.

While ethnographic studies of veterans, patterns of enrollment behavior, and prior work among traditional students motivated our intervention components, a key limitation of our study is the lack of baseline information documenting the nature and magnitude of the challenges that the intervention was designed to address. For example, it would have been informative to quantify the uncertainty and inaccuracy of service member awareness of differences in graduation rates and net costs at baseline, and if the intervention influenced this. From the perspective of study design, one key takeaway is that nudge interventions (particularly those that do not influence behavior) could be more informative if they better documented the baseline extent of (and potentially effect on) the challenges they are designed to address.

From a policy perspective, the results suggest the provision of personalized information, nudges, and assistance, which has previously been shown to be effective at addressing similar challenges among traditional student populations, at least in certain contexts, may simply be insufficient to overcome inertia in enrollment (many service members who enrolled while in the military continued at the same institution after the intervention) or the strong tendency to enroll at for-profit institutions. Earlier and/or more costly interventions, or more assertive policies may be necessary to improve postsecondary education and employment outcomes among veterans and other non-traditional student populations.

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